INCH-POUND
MIL-DTL-16377/7B(SH)
30 June 2014
SUPERSEDING
MIL-F-16377/7A(SH)
29 July 1981

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; T5 FLUORESCENT AND SSL EQUIVALENT, GENERAL LIGHTING, 16 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 70.3, 70.4, 78.2, 336, 336.1, 70.3SSL, 70.4SSL, 78.2SSL, 336SSL, AND 336.1SSL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

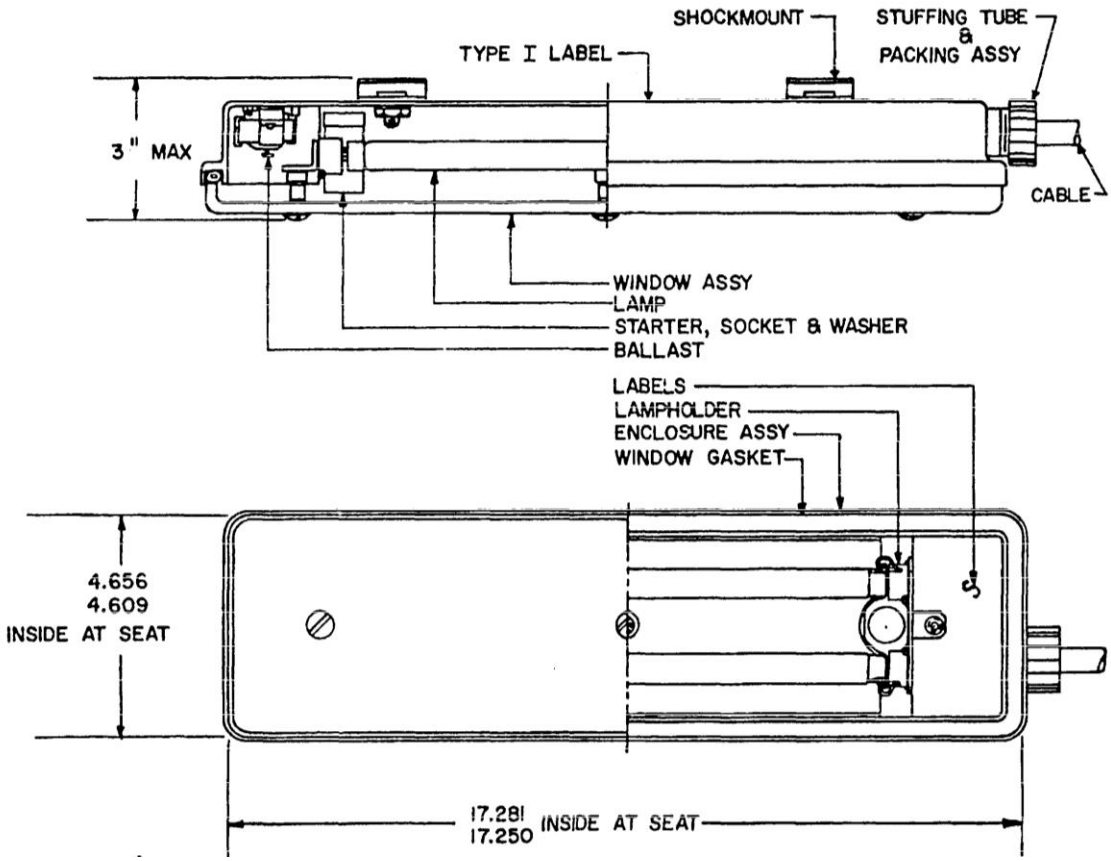


FIGURE 1. Typical view of symbols 70.3 and 336.

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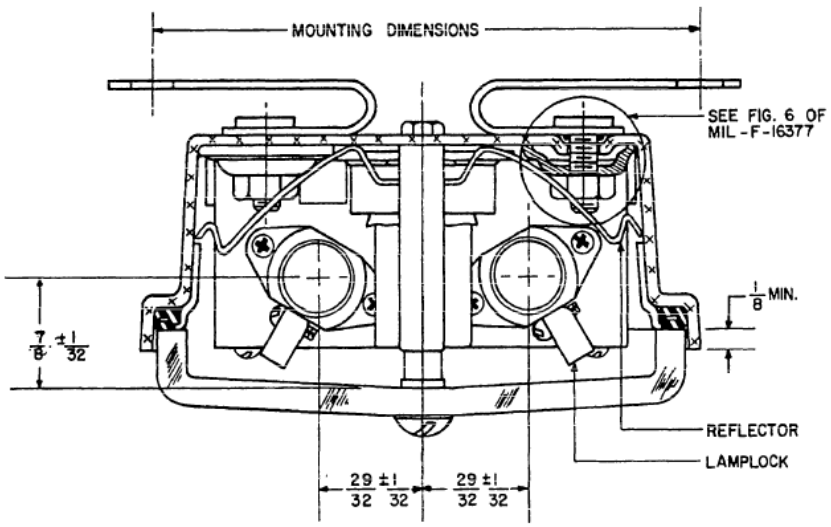


FIGURE 2. Typical section for symbols 70.3 and 336.

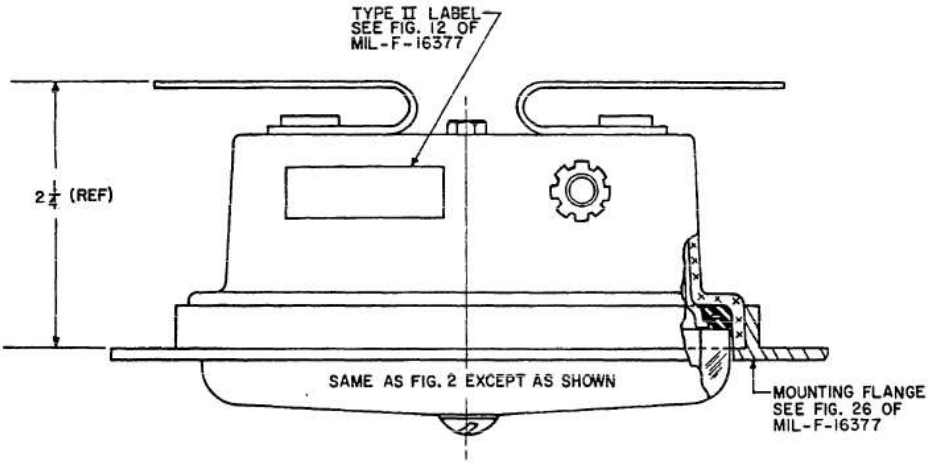
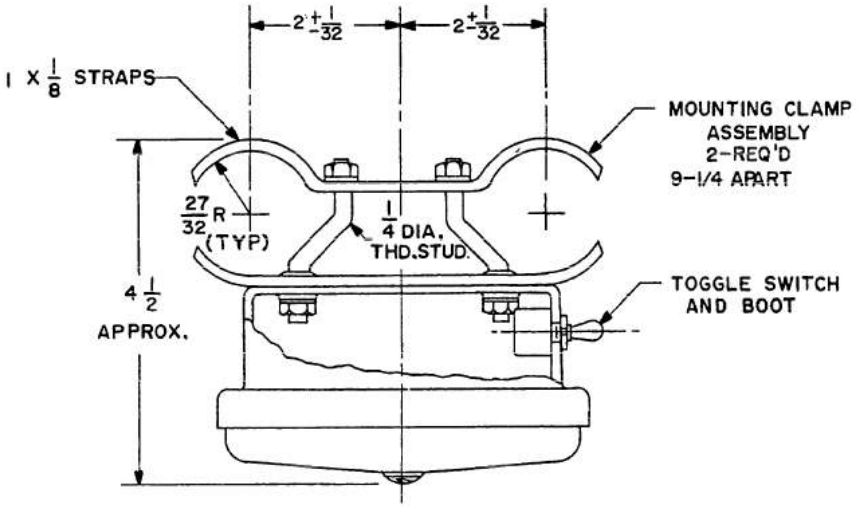


FIGURE 3. Typical section for symbols 70.4 and 336.1.

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SAME AS FIGURE 2, SYMBOL 336 EXCEPT THAT MOUNTING CLAMP ASSEMBLY AND TOGGLE SWITCH AND BOOT SHALL BE ADDED AND SHOCKMOUNTS OMITTED.

FIGURE 4. Typical section for symbol 78.2.

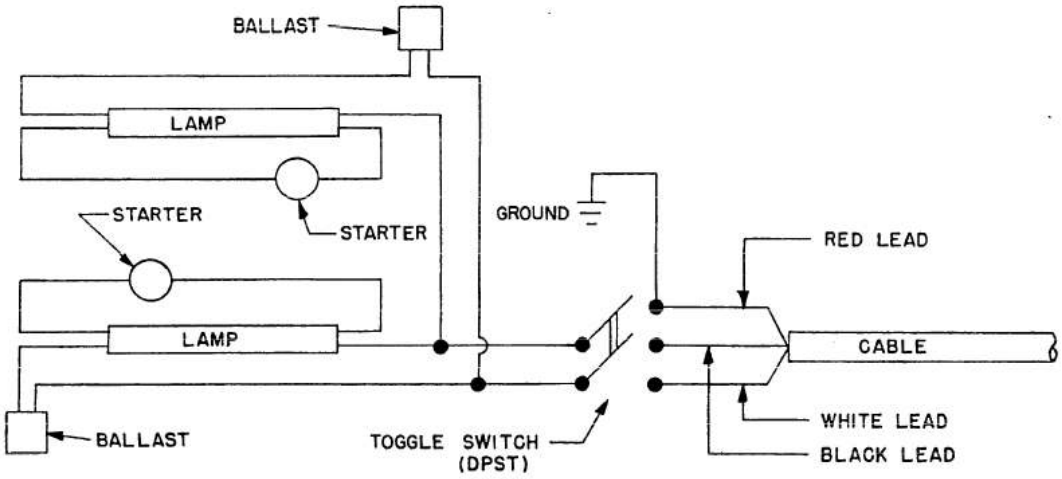
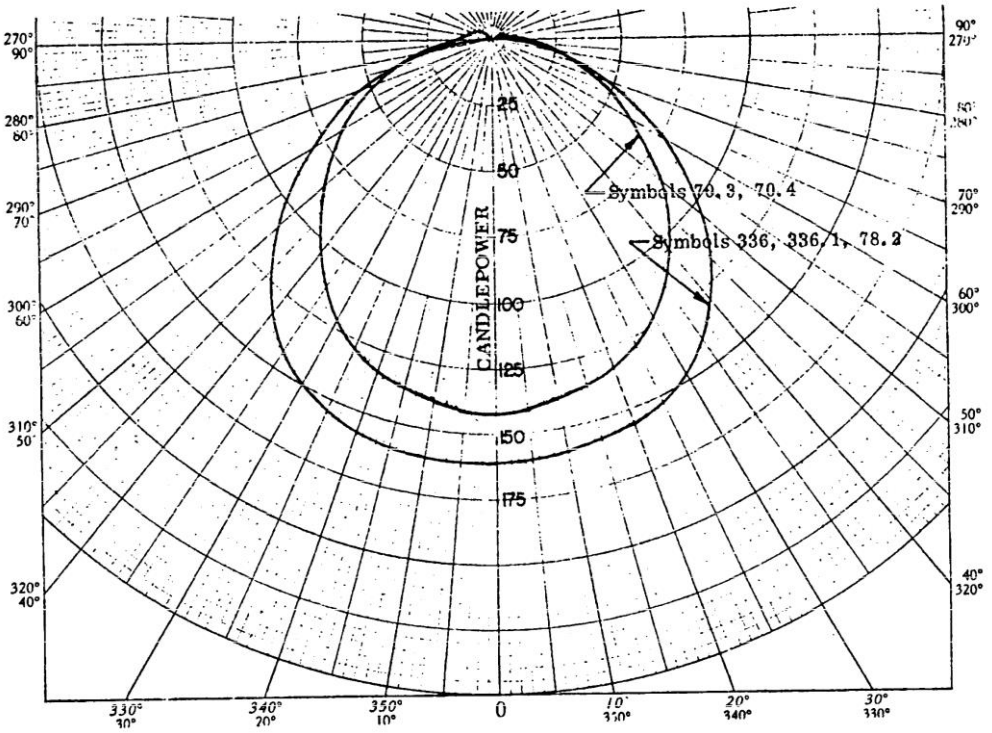


FIGURE 5. Wiring diagram for symbol 78.2.

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ANGLE	CANDLEPOWER SYMBOL NUMBER	
	70.3, 70.4	336, 336.1, 78.2
0	143	157
5	141	157
10	135	157
15	135	155
20	131	153
25	125	149
30	120	147
35	115	143
40	102	136
45	95	122
50	87	107
55	75	92
60	67	80

ANGLE	CANDLEPOWER SYMBOL NUMBER	
	70.3, 70.4	336, 336.1, 78.2
65	56	63
70	48	53
75	38	37
80	30	28
85	20	20
90	15	13
95	9	7
100	7	5
105	7	5
110	6	5
115	5	3
120	4	0
125	0	0

PHOTOMETRIC TEST DATA:

Distance:	10 feet.
Plane:	At right angles to long axis of fixture through geometric center of lamp.
Voltage:	115 volts, 60 hertz.
Ambient temperature:	75 °F (23.89 °C).
Lamps:	Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast:	Use ballast furnished with fixture.
Lamp correction:	Distribution curves based on 380 lumens per lamp.

FIGURE 6. Candlepower distribution curves.

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FLUORESCENT REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBER:

70.3: Complete lighting fixture assembly, standard mounting, with two 8-watt cool white fluorescent lamps and white diffusing window assembly.

70.4: Complete lighting fixture assembly, flush mounting, with two 8-watt cool white fluorescent lamps and white diffusing window assembly.

78.2: Complete lighting fixture assembly, with two 8-watt cool white fluorescent lamps and clear and white prismatic window assembly for reading and berthing table unit.

336: Complete lighting fixture assembly, standard mounting, with two 8-watt cool white fluorescent lamps and clear and white prismatic window assembly.

336.1: Complete lighting fixture assembly, flush mounting, with two 8-watt cool white fluorescent lamps and clear and white prismatic window assembly.

DESIGN:

Electrical rating:	16 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 1, 2, 3, and 4 and figure 10 of MIL-DTL-16377.
Photometric data:	See figure 6.
Lampholder spacing:	See figure 21 of MIL-DTL-16377.
Wiring diagram:	See figure 23 of MIL-DTL-16377 for Symbols 70.3, 70.4, 336, and 336.1 and figure 5 for Symbol 78.2.
Mounting dimensions:	
Symbols 70.3 and 336:	See figure 25 of MIL-DTL-16377.
Symbols 70.4 and 336.1:	See figure 26 of MIL-DTL-16377.
Window assembly:	
Symbols 70.3 and 70.4:	Military part number M16377/7-001, figure 27 of MIL-DTL-16377.
Symbols 78.2, 336, and 336.1:	Military part number M16377/7-002, figure 27 of MIL-DTL-16377.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Cable:	10 feet of Type LSTHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbols 70.3, 78.2, and 336:	5 pounds 10 ounces maximum.
Symbols 70.4 and 336.1:	6 pounds 12 ounces maximum.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 3:

a. Two Type I.
b. One Type II for Symbol 70.4 and 336.1 only.
c. One Type IV. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by a list of all standard parts shown in table I and window assemblies of table V by name and stock number only.

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SSL REQUIREMENTS:

CLASSIFICATION:

Type: III
Class: 2

SYMBOL NUMBER:

70.3SSL: Complete lighting fixture assembly, standard mounting, with LED luminaire and white diffusing window assembly.
70.4SSL: Complete lighting fixture assembly, flush mounting, with LED luminaire and white diffusing window assembly.
78.2SSL: Complete lighting fixture assembly, with LED luminaire and clear and white prismatic window assembly.
336SSL: Complete lighting fixture assembly, standard mounting, LED luminaire and clear and white prismatic window assembly.
336.1SSL: Complete lighting fixture assembly, flush mounting, with LED luminaire and clear and white prismatic window assembly.

DESIGN:

Electrical rating:	16 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 1, 2, 3, and 4 and figure 10 of MIL-DTL-16377.
Photometric data:	See figure 6 and 3.7.1 of MIL-DTL-16377.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.17 of MIL-DTL-16377.
Wiring:	See figure 23 of MIL-DTL-16377 for Symbols 70.3SSL, 70.4SSL, 336SSL, and 336.1SSL and figure 5 for Symbol 78.2SSL.
Mounting dimensions:	
Symbols 70.3SSL and 336SSL:	See figure 25 of MIL-DTL-16377.
Symbols 70.4SSL and 336.1SSL:	See figure 26 of MIL-DTL-16377.
Window assembly:	
Symbols 70.3SSL and 336SSL:	Military part number M16377/7-001, figure 27 of MIL-DTL-16377.
Symbols 70.4SSL and 336.1SSL:	Military part number M16377/7-002, figure 27 of MIL-DTL-16377.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Cable:	10 feet of Type LSTHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbols 70.3SSL, 78.2SSL, and 336SSL:	5 pounds 10 ounces maximum.
Symbols 70.4SSL and 336.1SSL:	6 pounds 12 ounces maximum.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 3:

a. Two Type I.
b. One Type II for Symbol 70.4SSL and 336.1SSL only.
c. One Type IV. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by:
   (1) A list of all standard parts shown in table I and window assemblies of table V by name and stock number only, if appropriate, or
   (2) A list of available replacement parts and stock numbers, or if unavailable, manufacturer's part numbers.

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EXAMINATION AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 and the following:

First article inspection:
- Complete fixtures: See table II for Type I and table III for Type III.
- Window assemblies: See table IV.

Conformance:
- Comparison inspection: Same as first article inspection.
- Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of military part numbers and general information is shown in table V.

TABLE I. Standard parts.

Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	ANSI/ANSLG C78.81	6240-00-299-5546	Industry number F8T5/CW
Lamp lock	M16377/40-001	6250-00-399-7039	
Lampholder	UL 496	6250-00-377-8590	SUPP. SA, FIG. SA2.1
Startersocket	UL 496	6250-00-295-2738	SUPP. SA, FIG. SA2.4
Starter	NEMA ANSI C78.180	9250-00-299-5962	Commercial designation: FS 5
Ballast	M16377/45-001	6250-00-295-2735	
Shock mount 1/	M16377/41-009 or M16377/41-010	6210-00-973-1747 6210-00-973-1748	See 3.5.16 of MIL-DTL-16377
Red filter tube	M16377/42-001	6210-00-399-6297	Not furnished, see 3.7.9 of MIL-DTL-16377
Blue filter tube	M16377/42-002	6210-00-503-2810	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (2.5% transmission)	M16377/42-014	6210-01-313-8170	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (1% transmission)	M16377/42-013	6210-01-351-3943	Not furnished, see 3.7.9 of MIL-DTL-16377
Switch, toggle, DPST	MS16569-1	5930-00-719-9659	For Symbol 78.2 and 78.2SSL only

NOTE:
1/ Shock mounts are not required for Symbol 78.2.

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TABLE II. First article and comparison inspection for complete type I fixtures. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/] Watertight
Shock	3.5.12	4.8.7	[2/, 3/]
Vibration	3.5.13	4.8.8	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/] Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Noise	3.5.14	4.8.9	[2/]
Grounding circuit	3.6.13.3	4.8.11	Symbol 78.2 only
Continuity of grounding	3.6.13.4	4.8.12	Symbol 78.2 excluded
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Symbols 70.3, 70.4, 336, and 336.1 shall withstand Grade A. Symbol 78.2 shall withstand Grade B.

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TABLE III. First article and comparison inspection for complete type III fixtures. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/] Watertight
Shock	3.5.12	4.8.7	[2/]
Vibration	3.5.13	4.8.8	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/] Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Noise	3.5.14	4.8.9	[2/]
Continuity of grounding	3.6.13.4	4.8.12	[2/]
Photometrics	3.7.4	4.8.6	
Brightness	3.7.4.1	4.8.6.4	
Chromaticity	3.7.4	4.8.6.5	
Correlated color temperature (CCT), white light only	3.7.4.1.1	4.8.6.6	
Lumen maintenance	3.7.11	4.8.20	
Fail-safe circuit design	3.17	4.8.24	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5	4.8.5	
Magnetic permeability	3.4.8	4.8.18	

NOTES:

[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.

[2/] In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE IV. First article and comparison inspection for window assemblies. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Optical	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Light output	3.7.1	4.8.6.1	Candlepower

NOTE:
1/ Requirements and test method paragraph numbers refer to MIL-DTL-16377.

TABLE V. Part number and general information.

Military Part Number M16377/7-	Item Name	National Stock Number (NSN)	Remarks
70.3	Complete fixture, Symbol 70.3	6210-01-595-3961	1/
70.4	Complete fixture, Symbol 70.4	6230-01-511-3184	1/
70.4S	Complete fixture, Symbol 70.4 with toggle switch	6210-01-484-6855	
78.2	Complete fixture, Symbol 78.2	6210-00-669-6526	1/
336	Complete fixture, Symbol 336	6210-01-149-4169	
336.1	Complete fixture, Symbol 336.1	6230-01-510-9625	
70.3SSL	Complete fixture, solid state, Symbol 70.3SSL		
70.4SSL	Complete fixture, solid state, Symbol 70.4SSL		
78.2SSL	Complete fixture, solid state, Symbol 78.2SSL		
336SSL	Complete fixture, solid state, Symbol 336SSL		
336.1SSL	Complete fixture, solid state, Symbol 336.1SSL		
001	Window assembly, white diffusing	6210-00-500-0742	2/
002	Window assembly, clear and white prismatic	6210-01-510-9785	

NOTES:
1/ Formerly covered on NAVSHIPS Drawing 9000-86401-74481 and MS17120.
2/ Formerly covered on MS16891.

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH

Review activity:
DLA – IS

Preparing activity:
Navy – SH
(Project 6210-2012-003)

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